manufacturing Rafale fuselage sections in India for Dassault's overseas orders. Indigenous content is planned to rise in phases to about 60 per cent. Separate agreements for key components, including the Rafale fuselage and Safran's M-88 engine, are in progress. Dassault has also established an MRO (maintenance, repair, overhaul) facility in India, with M-88 engine overhauls planned in Hyderabad.

The IAF's previous deal of 36 Rafales included IAF-specific enhancements, including the integration of an Israeli helmet-mounted display (HMD), radar warning receivers and low-band jammers. It also included a weapons package and a performance-based logistics agreement. The air-to-air missiles include Matra Magic II, MBDA MICA IR, and MBDA Meteor as well as Hammer bombs—used to destroy terror hubs in Pakistan during Op. Sindoor. A similar arrangement on integrating HMDs, EW suites and radar can be expected in the current deal too.

SOURCE CODE ISSUES

Then as now, a specific issue arises with France's refusal to share the Rafale's source code—computer programming that controls critical systems like the Thales AESA radar and the Modular Mission Computer (MMC)—citing security and strategic reasons. Though Dassault did help integrate Indian weapons like the Astra Mk1 air-to-air missiles and the SAAW (Smart Anti-Airfield Weapon), the lack of source code will force India to depend on the manufacturer for further integration of the jet with new weapons/ sensors. As a way out, military aviation experts say that India should aim for API (Application Programming Interface)-level access that would allow it to integrate specific software modules, weapons and sensors using software interfaces even without full access to the core source code.

THE JET STREAM LEAGUE

A LOOK AT THE LEADING FIGHTER AIRCRAFT OPERATED BY INDIA, CHINA AND PAKISTAN



INDIA

Total number: **570+**

Sukhoi Su-30MKI
No. : **270+**
Russian-origin multirole fighter manufactured in India under licence; several require upgrades

SEPECAT Jaguar
No. : **115-120**
British-made jet ideal for air support and ground attack; to be phased out by 2035

MiG-29
No. : **65**
Multirole Russian jet, upgraded to Mig-29 UPG; to be decommissioned by 2035

Dassault Rafale No. : **36**
French 4.5-generation multirole fighter, IAF's most advanced jet; 114 more to be acquired

Mirage 2000
No. : **54**
French multirole 4th generation fighter jet; to retire by 2035

Tejas LCA Mk1
No. : **32**
Indigenous supersonic multirole fighter

IN DEVELOPMENT:
LCA Mk 1A, Mk 2, 5th-generation AMCA

CHINA

Total Number: **1,400-1,500**

J-11 (variants J-11B/BS)
No. : Over **350**
4th-generation multirole fighter

₹3.25 LAKH CRORE
Estimated cost of purchasing 114 Rafales, including 18 in fly-away condition and 96 made in India

Squadron Leader Vijainder K. Thakur (retd), a military analyst, says, "Access to the platform's source code should be limited to API-level integration, sufficient to deploy Indian weapons. Transfer of Technology (ToT) should be delinked from weapon acquisition." He clarifies that tying weapons acquisition to ToT increases costs and constrains the IAF's ability to procure munitions, hurting operational readiness.

However, some experts believe that a big induction of Rafales would hamper India's own fighter jet programme, as much of IAF's budget would be diverted for it. Moreover, delivery timelines of the Rafales are expected to be the same (early 2030s onwards) as the Tejas Mk2, also a 4.5 generation fighter.

Some others have issues with the 'made in India' part. Despite manufacturing—focused largely on assembly and manufacturing under transferred knowhow—nearly 850 MiG-21 aircraft domestically, India did not develop an independent aerospace ecosystem